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Uniform Plane Waves

GATE ECE Electromagnetics · Previous Year Questions with Answer Key

78	41	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1 25 questions

Q1. GATE ECE 2026

The electric field of a monochromatic plane wave travelling in a lossless isotropic and homogenous medium is given by

$\vec{E}(z,t) = E_0[\hat{x}\cos(\omega t - kz) + \hat{y}\sin(\omega t - kz)]$

in a right-handed orthogonal co-ordinate system.

Which of the following is the correct polarization of the electromagnetic wave?

(A) Right-handed circularly polarized

(B) Left-handed circularly polarized

(C) Linearly polarized

(D) Linearly polarized with -45° angle to $\hat{x}$

Open solution

Q2. GATE ECE 2023

Consider a narrow band signal, propagating in a lossless dielectric medium $(\varepsilon_r = 4, \mu_r = 1)$, with phase velocity v_p and group velocity v_g. Which of the following statement is true? (c is the velocity of light in vacuum.)

(A) $v_p > c, v_g > c$ (B) $v_p < c, v_g > c$ (C) $v_p > c, v_g < c$ (D) $v_p < c, v_g < c$

Open solution

Q3. GATE ECE 2022

Consider the following wave equation,

$\frac{\partial^2 f(x,t)}{\partial t^2} = 10000\frac{\partial^2 f(x,t)}{\partial x^2}$

Which of the given options is/are solution(s) to the given wave equation?

(A) $f(x,t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$

(B) $f(x,t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$

(C) $f(x,t) = e^{-(x-100t)} + \sin(x+100t)$

(D) $f(x,t) = e^{j100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Open solution

Q4. GATE ECE 2021

Consider a rectangular coordinate system (x, y, z) with unit vectors a_x, a_y and a_z. A plane wave traveling in the region $z \geq 0$ with electric field vector $E = 10\cos\left(2 \times 10^8 t + \beta z\right)a_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity $\varepsilon = \varepsilon_0$ permeability $\mu = 4\mu_0$, where $\varepsilon_0 = 8.85 \times 10^{-12}\mathrm{F/m}$ and $\mu_0 = 4\pi \times 10^{-7}\mathrm{H/m}$). The value of the reflection coefficient is

(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Open solution

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Q5. GATE ECE 2016 Set 2

Let the electric field vector of a plane electromagnetic wave propagating in a homogenous medium be expressed as $E = \hat{x}E_x\, e^{-j(wt-\beta z)}$, , where the propagation constant β is a function of the angular frequency ω. Assume that $\beta(\omega)$ and E_x are known and are real. From the information available, which one of the following CANNOT be determined?

(A) The type of polarization of the wave.
(B) The group velocity of the wave.
(C) The phase velocity of the wave.
(D) The power flux through the z = 0 plane.

Open solution

Q6. GATE ECE 2016 Set 3

If a right-handed circularly polarized wave is incident normally on a plane perfect conductor, then the reflected wave will be

(A) right-handed circularly polarized
(B) left-handed circularly polarized
(C) elliptically polarized with a tilt angle of 450
(D) horizontally polarized

Open solution

Q7. GATE ECE 2015 Set 2

The electric field of a uniform plane electromagnetic wave is

$$\vec{E} = \left(\overrightarrow{a_x} + j4\overrightarrow{a_y}\right) \exp\left[j\left(2\pi \times 10^7 t - 0.2z\right)\right]$$

The polarization of the wave is

(A) right handed circular
(B) right handed elliptical
(C) left handed circular
(D) left handed elliptical

Open solution

Q8. GATE ECE 2015 Set 1

The electric field component of a plane wave traveling in a lossless dielectric medium is given by

$$\overrightarrow{E}(z,t) = \hat{a}_y 2\cos\left(10^8 t - \frac{z}{\sqrt{2}}\right) V/m.$$

The wavelength)in m) for the wave is

Open solution

Q9. GATE ECE 2014 Set 1

A two - port network has scattering parameters given by $[S] = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$.

If the port - 2 of the two - port is short circuited, the s_{11} parameter for the resultant one - port network is

(A) $\frac{s_{11}-s_{11}s_{22}+s_{12}s_{21}}{1+s_{22}}$
(B) $\frac{s_{11}+s_{11}s_{22}-s_{12}s_{21}}{1+s_{22}}$
(C) $\frac{s_{11}+s_{11}s_{22}+s_{12}s_{21}}{1-s_{22}}$
(D) $\frac{s_{11}-s_{11}s_{22}+s_{12}s_{21}}{1-s_{22}}$

Open solution

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Q10. GATE ECE 2014 Set 2

Which one of the following field patterns represents a TEM wave traveling in the positive x direction?

Open solution

(A) $E = +8\hat{y},\ H = -4\hat{z}$

(B) $E = -2\hat{y},\ H = -3\hat{z}$

(C) $E = +2\hat{z},\ H = +2\hat{y}$

(D) $E = -3\hat{y},\ H = +4\hat{z}$

Q11. GATE ECE 2012

The electric field of a uniform plane electromagnetic wave in free spce, along the positive x direction, is given by $\vec{E} = 10\left(\hat{a}_y + j\hat{a}_z\right)e^{-j25x}$. The frequency and polarization of the wave respectively are

Open solution

(A) $1.2GHz$ and left circular

(B) $4Hz$ and left circular

(C) $1.2GHz$ and right circular

(D) $4Hz$ and right circular

Q12. GATE ECE 2012

A plane wave propagating in air with $\vec{E} = \left(8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z\right)e^{j(\omega t+3x-4y)}\,V/m$ is incident on a perfectly conducting slab positioned at $x \leq 0$. The $\overrightarrow{E}$ field of the reflected wave is

Open solution

(A) $\left(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z\right)e^{j(\omega t+3x+4y)}\,V/m$

(B) $\left(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z\right)e^{j(\omega t+3x+4y)}\,V/m$

(C) $\left(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z\right)e^{j(\omega t-3x-4y)}\,V/m$

(D) $\left(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z\right)e^{j(\omega t-3x-4y)}\,V/m$

Q13. GATE ECE 2007

A plane wave of wavelength λ is traveling in a direction making an angle 30° with positive x-axis and 90° with positiv y-axis. The $\overrightarrow{E}$ field of the plane wave can be represented as (E_0 is a constant)

Open solution

(A) $\vec{E} = \hat{y}\ E_0 e^{j\left(\omega t - \frac{\sqrt{3}\pi}{\lambda}x - \frac{\pi}{\lambda}z\right)}$

(B) $\vec{E} = \hat{y}\ E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x - \frac{\sqrt{3}\pi}{\lambda}z\right)}$

(C) $\vec{E} = \hat{y}\ E_0 e^{j\left(\omega t + \frac{\sqrt{3}\pi}{\lambda}x + \frac{\pi}{\lambda}z\right)}$

(D) $\vec{E} = \hat{y}\ E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x + \frac{\sqrt{3}\pi}{\lambda}z\right)}$

Q14. GATE ECE 2006

The electric field of an electromagnetic wave propagating in the positive z-direction is given by

$$E = \hat{a}_x \sin(\omega t - \beta z) + \hat{a}_y \sin(\omega t - \beta z + \pi/2)$$

Open solution

The wave is

(A) linearly polarized in the z-direction

(B) elliptically polarized

(C) left-hand circularly polarized

(D) right-hand circularly polarized

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Q15. GATE ECE 2005

The magnetic field intensity vector of a plane wave is given by
$\overline{H}(x, y, z, t) = 10 \sin(50000t + 0.004x + 30)\hat{a}_y$

Where $\hat{a}_y$ denotes the unit vector in y direction. The wave is propagating with a phase velocity

(A) $5 \times 10^4\ m/s$ (B) $-3 \times 10^8\ m/s$ (C) $-1.25 \times 10^7\ m/s$ (D) $3 \times 10^8\ m/s$

Open solution

Q16. GATE ECE 2003

The depth of penetration of electromagnetic wave in a medium having conductivity σ at a frequency of 1 KHz is 25 cm. The depth of penetration at a frequency of 4 KHz will be

(A) 6.25 cm (B) 12.50 cm (C) 50.00 cm (D) 100.00 cm

Open solution

Q17. GATE ECE 2001

If a plane electromagnetic wave satisfies the equation $\frac{\partial^2 E_x}{\partial z^2} = c^2 \frac{\partial^2 E_x}{\partial t^2}$, the wave propagates in the

(A) x-direction

(B) z-direction

(C) y-direction

(D) x z plane at an angle of 45° between the x and z directions

Open solution

Q18. GATE ECE 2000

A TEM wave is incident normally upon a perfect conductor. The E and H fields at the boundary will be respectively

(A) minimum and minimum (B) maximum and maximum

(C) minimum and maximum (D) maximum and minimum

Open solution

Q19. GATE ECE 1998

The intrinsic impedance of copper at high frequencies is

(A) purely resistive. (B) purely inductive.

(C) complex with a capacitive component. (D) complex with an inductive component.

Open solution

Q20. GATE ECE 1998

The wavelength of a wave with propagation constant $(0.1\pi + j0.2\pi)m^{-1}$ is

(A) $\frac{2}{\sqrt{0.05}}\ m$ (B) $10\,m$ (C) $20\,m$ (D) $30\,m$

Open solution

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Q21. GATE ECE 1998

The depth of penetration of a wave in a lossy dielectric increases with increasing

(A) conductivity (B) permeability (C) wavelength (D) permittivity

Open solution

Q22. GATE ECE 1998

The polarization of a wave with electric field vector $\overrightarrow{E} = E_0\, e^{j(\omega t - \beta z)} \left(\overrightarrow{a_x} + \overrightarrow{a_y}\right)$ is

(A) linear
(B) elliptical
(C) left hand circular
(D) right hand circular

Open solution

Q23. GATE ECE 1995

The intrinsic impedance of a lossy dielectric medium is given by

(A) $\frac{j\omega\mu}{\sigma}$
(B) $\frac{j\omega\in}{\mu}$
(C) $\sqrt{\frac{j\omega\mu}{\sigma+j\omega\in}}$
(D) $\sqrt{\frac{\mu}{\in}}$

Open solution

Q24. GATE ECE 1995

Copper behaves as a

(A) conductor always.
(B) conductor or dielectric depending on the applied electric field strength.
(C) conductor or dielectric depending on the frequency.
(D) conductor or dielectric depending on the electric current density.

Open solution

Q25. GATE ECE 1994

A plane electromagnetic wave traveling along the +z-direction, has its electric field given by E_x = 2 cos (ωt) and E_y = 2 cos $(\omega t + 90^\circ)$ the wave is

(A) linearly polarized
(B) right circularly polarized
(C) left circularly polarized
(D) elliptically polarized

Open solution

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Marks 2 41 questions

Q26. GATE ECE 2024

A uniform plane wave with electric field $\vec{E}(x) = A_y \hat{a}_y e^{-j\frac{2\pi x}{3}}$ V/m is travelling in the air (relative permittivity, $\dot{o}_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____.

Open solution

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$ (C) -6 (D) -3

Q27. GATE ECE 2023

The electric field of a plane electromagnetic wave is

$E = a_x C_{1x} \cos(\omega t - \beta z) + a_y C_{1y} \cos(\omega t - \beta z + \theta)$ V/m.

Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?

Open solution

(A) $C_{1x} = 1, C_{1y} = 1, \theta = \pi/4$ (B) $C_{1x} = 2, C_{1y} = 1, \theta = \pi/2$

(C) $C_{1x} = 1, C_{1y} = 2, \theta = 3\pi/2$ (D) $C_{1x} = 2, C_{1y} = 1, \theta = 3\pi/4$

Q28. GATE ECE 2023

A transparent dielectric coating is applied to glass ($\varepsilon_r = 4, \mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75\ \mu$m). The minimum thickness of the dielectric coating, in μm, that can be used is _____ (rounded off to two decimal places).

Open solution

Q29. GATE ECE 2021

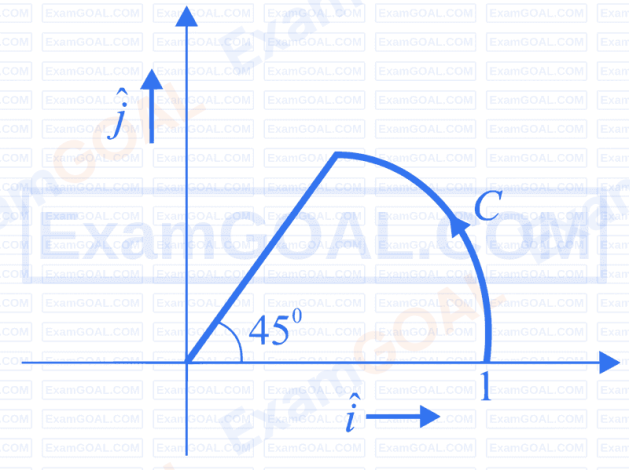
For a vector field $\overrightarrow{\mathbf{D}} = \rho \cos^2 \phi \hat{\mathbf{a}}_{\boldsymbol{\rho}} + z^2 \sin^2 \phi \hat{\mathbf{a}}_\phi$ in a cylindrical coordinate system (ρ, ϕ, z) with unit vector $\hat{\boldsymbol{a}}_\rho, \hat{\boldsymbol{a}}_\phi$ and $\hat{\boldsymbol{a}}_z$, the net flux of $\overrightarrow{\mathbf{D}}$ leaving the closed surface of the cylinder $(\rho = 3, 0 \leq z \leq 2)$ (Round off to 2 decimal places) is ____ .

Open solution

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Q30. GATE ECE 2021

The vector function $F(r) = -x\hat{i} + y\hat{j}$ is defined over a circular arc C shown in the figure,



The line integral of $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$ is

(A) 1/2 (B) 1/4 (C) 1/6 (D) 1/3

Open solution

Q31. GATE ECE 2021

Consider the vector field $\overline{\mathbf{F}} = \hat{\mathbf{a}}_\mathbf{x}(4y - c_1 z) + \hat{\mathbf{a}}_\mathbf{y}(4x + 2z) + \hat{\mathbf{a}}_\mathbf{z}(2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors $\hat{\mathbf{a}}_\mathbf{x}, \hat{\mathbf{a}}_\mathbf{y}, \hat{\mathbf{a}}_\mathbf{z}$. If the field $\mathbf{F}$ is irrotational (conservative), then the constant c_1 (in integer) is ____ .

Open solution

Q32. GATE ECE 2020

The magnetic field of a uniform plane wave in vacuum is given by

$\vec{H}(x, y, z, t) = \left(\hat{a}_x + 2\hat{a}_y + b\hat{a}_z\right) \cos(\omega t + 3x - y - z)$.

The value of b is ____ .

Open solution

Q33. GATE ECE 2017 Set 1

The expression for an electric field in free space is $E = E_0\left(\hat{x} + \hat{y} + j2\hat{z}\right)e^{-j(\omega t - kx + ky)}$, where $x,\ y,\ z$ represent the spatial coordinates, t represents time, and $\omega,\ k$ are contants. This electric field

(A) does not represent a plane wave.

(B) represents a circular polarized plane wave propagating normal to the z-axis.

(C) represents an elliptically polarized plane wave propagating along x-y plane.

(D) represents a linearly polarized plane wave

Open solution

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Q34. GATE ECE 2016 Set 1

Open solution

The electric field of a uniform plane wave travelling along the negative z direction is given by the following equation:

$$\overrightarrow{E_w^i} = (\hat{a}_x + j\hat{a}_y)E_0 e^{jkz}$$

This wave is incident upon a receiving antenna placed at the origin and whose radiated electric field towards the incident wave is given by the following equation:

$$\overrightarrow{E_a} = (\hat{a}_x + 2\hat{a}_y)E_1 \frac{1}{r} e^{-jkr}$$

The polarization of the incident wave, the polarization of the antenna and losses due to the polarization mismatch are, respectively,

(A) Linear, Circular (clockwise), $-5dB$

(B) Circular (clockwise), Linear, $-5dB$

(C) Circular (clockwise), Linear, $-3dB$

(D) Circular (anti clockwise), Linear, $-3dB$

Q35. GATE ECE 2015 Set 2

Open solution

The electric field intensity of a plane wave propagating in a lossless non-magnetic medium is given by the following expression

$$\overrightarrow{E}(z,t) = \hat{a}_x 5\cos\left(2\pi \times 10^9 t + \beta z\right)$$

$$+\hat{a}_y 3\cos\left(2\pi \times 10^9 t + \beta z - \frac{\pi}{2}\right)$$

The type of the polarization is

(A) Right hand circular

(B) Left hand elliptical

(C) Right hand elliptical

(D) Linear

Q36. GATE ECE 2015 Set 1

Open solution

The electric field intensity of a plane wave traveling in free space is given by the following expression

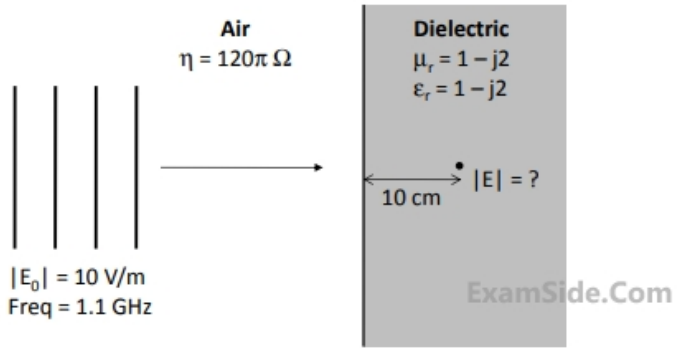
$E(x,t) = \hat{a}_y 24\pi \ \cos(\omega t - k_0 x) \ \ (V/m)$. In this field, consider a square area $10cm \times 10cm$ on a plane $x + y = 1$. The total time-averaged power $(inmW)$ passing through the square area is _____.

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Q37. GATE ECE 2015 Set 1

Open solution

Consider a uniform plane wave with amplitude (E_0) of $10\ V/m$ and $1.1GHz$ frequency travelling in air, and incident normally on a dielectric medium with complex relative permittivity (ε_r) and permeability (μ_r) as shown in the figure.



The magnitude of the transmitted electric field component (in V/m) after it has travelled a distance of 10 cm inside the dielectric region is _____.

Q38. GATE ECE 2014 Set 1

Open solution

In spherical coordinates, let $\hat{a}_\theta$, $\hat{a}_\phi$ denote unit vectors along the θ, ϕdirections.

$$E = \frac{100}{r}\sin\theta\cos(\omega t - \beta r)\hat{a}_\theta\ V/m$$

and

$$H = \frac{0.265}{r}\sin\theta\cos(\omega t - \beta r)\hat{a}_\phi\ A/m$$

Represent the electric and magnetic field components of the EM wave at large distances r from a dipole antenna, in free space. The average power (W) crossing the hemispherical shell located at $r = 1\ km, 0 \le \theta \le \pi/2$ _____.

Q39. GATE ECE 2014 Set 2

Open solution

If the electric field of a plane wave is

$$\vec{E}(z,t) = \hat{x}3\cos(\omega t - kz + 30^\circ) - \hat{y}4\sin(\omega t - kz + 45^\circ)\,(mV/m)$$

The polarization state of the plane wave is

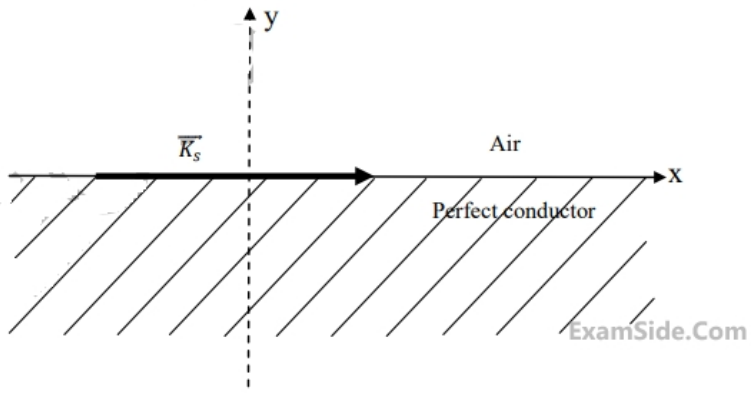
(A) left elliptical (B) left circular (C) right elliptical (D) right circular

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Q40. GATE ECE 2014 Set 3

Open solution

A region shown below contains a perfect conducting half-space and air. The surface current $\vec{K}_s$ on the surface of the perfect conductor is $\vec{K}_s = \hat{x}\,2$ amperes per meter. The tangential $\vec{H}$ field in the air just above the perfect conductor is



(A) $(\hat{x}+\hat{z})2$ amperes per meter

(B) $\hat{x}2$ amperes per meter

(C) $-\hat{z}2$ amperes per meter

(D) $\hat{z}2$ amperes per meter

Q41. GATE ECE 2014 Set 3

Open solution

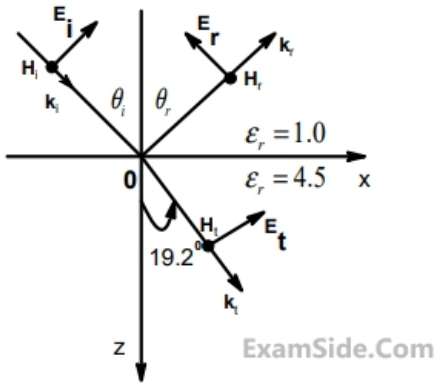
Assume that a plane wave in air with an electric field $\vec{E} = 10\cos\left(\omega t - 3x - \sqrt{3}z\right)\hat{a}_y \ V/m$ is incident on a non-magnetic dielectric slab of relative permittivity 3 which covers the region $z > 0$. The angle of transmission in the dielectric slab is _____ degrees.

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Q42. GATE ECE 2013

Open solution

A monochromatic plane wave of wavelength $\lambda = 600$ is propagating in the direction as shown in the figure below. $\overrightarrow{E_i}$, $\overrightarrow{E_r}$ and $\overrightarrow{E_t}$ denote incident, reflected, and transmitted electric field vectors associated with the wave.



The angle of incidence θ_i and the expression for $\overrightarrow{E_i}$ are

(A) $60^\circ \ and \ \frac{E_0}{\sqrt{2}}\left(\hat{a}_x - \hat{a}_z\right)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}} \ V/m$

(B) $45^\circ \ and \ \frac{E_0}{\sqrt{2}}\left(\hat{a}_x - \hat{a}_z\right)e^{-j\frac{\pi \times 10^4 z}{3}} \ V/m$

(C) $45^\circ \ and \ \frac{E_0}{\sqrt{2}}\left(\hat{a}_x - \hat{a}_z\right)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}} \ V/m$

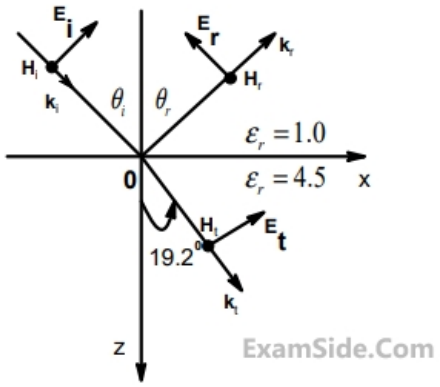
(D) $60^\circ \ and \ \frac{E_0}{\sqrt{2}}\left(\hat{a}_x - \hat{a}_z\right)e^{-j\frac{\pi \times 10^4 z}{3}} \ V/m$

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Q43. GATE ECE 2013

Open solution

A monochromatic plane wave of wavelength $\lambda = 600$ is propagating in the direction as shown in the figure below. $\overrightarrow{E_i}$, $\overrightarrow{E_r}$ and $\overrightarrow{E_t}$ denote incident, reflected, and transmitted electric field vectors associated with the wave.



The expression for $\overrightarrow{E_r}$ is

(A) $0.23\frac{E_0}{\sqrt{2}}\left(\widehat{a}_x - \widehat{a}_z\right)e^{-j\frac{\pi\times10^4(x-z)}{3\sqrt{2}}}\ V/m$

(B) $-\frac{E_0}{\sqrt{2}}\left(\widehat{a}_x - \widehat{a}_z\right)e^{-j\frac{\pi\times10^4 z}{3}}\ V/m$

(C) $0.44\frac{E_0}{\sqrt{2}}\left(\widehat{a}_x - \widehat{a}_z\right)e^{-j\frac{\pi\times10^4(x-z)}{3\sqrt{2}}}\ V/m$

(D) $\frac{E_0}{\sqrt{2}}\left(\widehat{a}_x - \widehat{a}_z\right)e^{-j\frac{\pi\times10^4(x+z)}{3}}\ V/m$

Q44. GATE ECE 2011

Open solution

The electric and magnetic fields for a TEM wave of frequency $14GHz$ in a homogeneous medium of relative permittivity ε_r and relative permeability $\mu_r = 1$ are given by

$$\overrightarrow{E} = E_p\ e^{j(\omega t - 280\pi y)}\ \widehat{u}_z\ V/m$$

$$\overrightarrow{H} = \ 3\ e^{j(\omega\, t - 280\ \pi\, y)}\ \widehat{u}_{\,x}\ A/m$$

Assuming the speed of light in free space to be $3\ \times 10^8\ m/s$, the intrinsic impedance of free space to be $120\ \pi$, the relative permittivity ε_r of the medium and the electric field amplitude E_p are

(A) $\varepsilon_r = 3,\ E_p = 120\,\pi$

(B) $\varepsilon_r = 3,\ E_p = 360\,\pi$

(C) $\varepsilon_r = 9,\ E_p = 360\,\pi$

(D) $\varepsilon_r = 9,\ E_p = 120\,\pi$

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Q45. GATE ECE 2010

A plane wave having the electric field component $\overrightarrow{E_i} = 24\ \cos\ (3 \times 10^8\, t - \beta y)\widehat{a}_z\ V/m$ and traveling in free space is incident normally on a lossless medium with $\mu = \mu_0$ and $\varepsilon = 9\ \varepsilon_0$, which occupies the region $y \geq 0$. The reflected magnetic field component is given by

Open solution

(A) $\frac{1}{10\pi} \cos\ (3 \times 10^8 t + y)\hat{a}_{\ x}\, A/m$

(B) $\frac{1}{20\pi} \cos\ (3 \times 10^8 t + y)\hat{a}_{\ x}\, A/m$

(C) $-\frac{1}{20\pi} \cos\ (3 \times 10^8 t + y)\hat{a}_{\ x}\, A/m$

(D) $-\frac{1}{10\pi} \cos\ (3 \times 10^8 t + y)\hat{a}_{\ x}\, A/m$

Q46. GATE ECE 2008

A uniform plane wave in the free space is normally incident on an infinitely thick dielectric slab (dielectric constant $\varepsilon_r = 9$). The magnitude of the reflection coefficient is

Open solution

(A) 0 (B) 0.3 (C) 0.5 (D) 0.8

Q47. GATE ECE 2007

The $\overrightarrow{H}$ field (in A/m) of a plane wave propagating in free space is given by

$$\overrightarrow{H} = \widehat{x}\frac{5\sqrt{3}}{\eta_0}\cos(\omega t - \beta z) + \hat{y}\frac{5}{\eta_0}\sin\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

Open solution

The time average power flow density in Watts is

(A) $\frac{\eta_0}{100}$ (B) $\frac{100}{\eta_0}$ (C) $50\,{\eta_0}^2$ (D) $\frac{50}{\eta_0}$

Q48. GATE ECE 2006

A medium of relative permittivityb $\varepsilon_r = 2$ forms an interface with free-space. A point source of electromagnetic energy is located in the medium at a depth of 1 meter from the interface. Due to the total internal reflection, the transmitted beam has a circular cross-section over the interface. The area of the beam cross-section at the interface is given by

Open solution

(A) $2\pi\, m^2$ (B) $\pi^2\, m^2$ (C) $\pi/2\, m^2$ (D) $\pi\, m^2$

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Q49. GATE ECE 2006

A medium is divided into regions I and II about $x = 0$ plane, as shown in the Fig. below. An electromagnetic wave with electric field $\vec{E_1} = 4\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ is incident normally on the interface form region-I . The electric field E_2 in region-II at the interface is

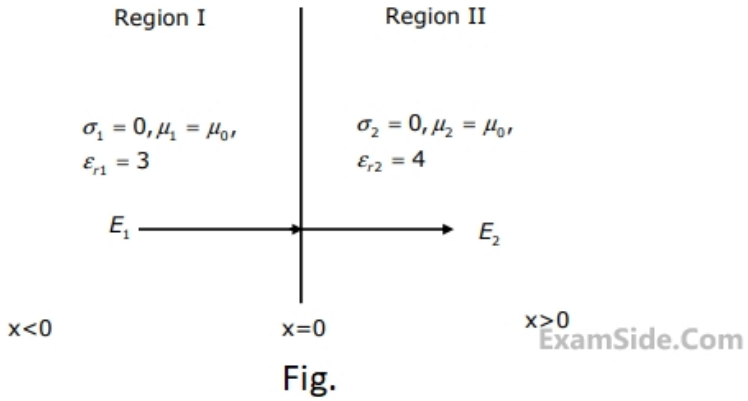


Fig.

Open solution

(A) $E_2 = E_1$
(B) $4\,\hat{a}_x + 0.75\,\hat{a}_y - 1.25\,\hat{a}_z$
(C) $3\,\hat{a}_x + 3\,\hat{a}_y + 5\,\hat{a}_z$
(D) $-3\,\hat{a}_x + 3\,\hat{a}_y + 5\,\hat{a}_z$

Q50. GATE ECE 2006

When a plane wave traveling in free-space is incident normally on a medium having $\varepsilon_r = 4.0$, the fraction of power transmitted into the medium is given by

(A) 8/9 (B) 1/2 (C) 1/3 (D) 5/6

Open solution

Q51. GATE ECE 2004

A plane electromagnetic wave propagating in free space in incident normally on a large slab of loss-less, non-magnetic, dielectric material with $\varepsilon > \varepsilon_0$. Maxima and minima are observed when the electric field is measured in front of the slab. The maximum electric field is found to be 5 times the minimum field. The intrinsic impedance of the medium should be

(A) $120\,\pi\,\Omega$ (B) $60\,\pi\,\Omega$ (C) $600\,\pi\,\Omega$ (D) $24\,\pi\,\Omega$

Open solution

Q52. GATE ECE 2003

Medium 1 has the electrical permittivity $\varepsilon_1 = 1.5\ \varepsilon_0\ F/m$ and occupies the region to left of $x = 0$ plane. Medium 2 has the electrical permittivity $\varepsilon_2 = 2.5\ \varepsilon_0\ F/m$ and occupies the region to the right of $x = 0$ plane. If E_1 in medium 1 is $E_1 = (2\,u_x - 3\,u_y + 1\,u_z)$ volt/m, then E_2 in medium 2 is

Open solution

(A) $(2.0\ u_x - 7.5\ u_y + 2.5\ u_z)$ volt/m
(B) $(2.0\ u_x - 2.0\ u_y + 0.6\ u_z)$ volt/m
(C) $(1.2\ u_x - 3.0\ u_y + 1.0\ u_z)$ volt/m
(D) $(1.2\ u_x - 2.0\ u_y + 0.6\ u_z)$ volt/m

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Q53. GATE ECE 2003

A uniform plane wave traveling in air is incident on the plane boundary between air and another dielectric medium with $\varepsilon_r = 4$. The reflection coefficient for the normal incidence, is

(A) zero (B) $0.5\angle 180^\circ$ (C) $0.333\angle 0^\circ$ (D) $0.333\angle 180^\circ$

Open solution

Q54. GATE ECE 2003

If the electric field intensity associated with a uniform plane electromagnetic wave traveling in a perfect dielectric medium is given by
$E(z, t) = 10 \cos(2\pi \times 10^7 t - 0.1 \pi z)$ volt/m, the velocity of the traveling wave is

(A) 3.00×10^8 m/sec (B) 2.00×10^8 m/sec
(C) 6.28×10^7 m/sec (D) 2.00×10^7 m/sec

Open solution

Q55. GATE ECE 2002

A plane wave is characterized by

$$\vec{E} = \left(0.5\hat{x} + \hat{y}e^{j\pi/2}\right)e^{j\omega t - jkz}.$$

This wave is

(A) linearly polarized (B) circularly polarized
(C) elliptically polarized (D) un polarized

Open solution

Q56. GATE ECE 2002

Distilled water at $25^\circ C$ is characterized by $\sigma = 1.7 \times 10^{-4}$ mho/m and $\in = 78\in_0$ at a frequency of $3GHz$. Its loss tangent $\tan\delta$ is

(A) 1.3×10^{-5} (B) 1.3×10^{-3}
(C) $1.7 \times 10^{-4}/78$ (D) $1.7 \times 10^{-4}(78\in_0)$

Open solution

Q57. GATE ECE 2001

A uniform plane electromagnetic wave incident normally on a plane surface of a dielectric material is reflected with a VSWR of 3. What is the percentage of incident power that is reflected?

(A) 10% (B) 25% (C) 50% (D) 75%

Open solution

Q58. GATE ECE 2000

A uniform plane wave in air impinges at 45° angle on a lossless dielectric material with dielectric constant ε_r. The transmitted wave propagates in a 30° direction with respect to the normal. The value of ε_r is

(A) 1.5 (B) $\sqrt{1.5}$ (C) 2 (D) $\sqrt{2}$

Open solution

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Q59. GATE ECE 1996

A uniform plane wave in air is normally incident on an infinitely thick slab. If the reflective index of the glass slab is 1.5, then the percentage of the incident power that is reflected from the air-glass interface is

Open solution

(A) 0% (B) 4% (C) 20% (D) 100%

Q60. GATE ECE 1996

Some unknown material has a conductivity of $10^6\ mho/m$ and a permeability of $4\pi \times 10^{-7}\ \ H/m$. The skin depth for the material at $1GHz$ is

Open solution

(A) $15.9\ \mu m$ (B) $20.9\ \mu m$ (C) $25.9\ \mu m$ (D) $30.9\ \mu m$

Q61. GATE ECE 1993

A material is described by the following electrical parameters at a frequency of 10 GHz is $\sigma = 10^6$ mho/m, $\mu = \mu_0$, and $\in /\in_0 = 10$. The material at this frequency is considered to be $\left(\in_0 = \frac{1}{36\ \pi} \times 10^{-9}\ F/m\right)$

Open solution

(A) a good conductor

(B) a good dielectric

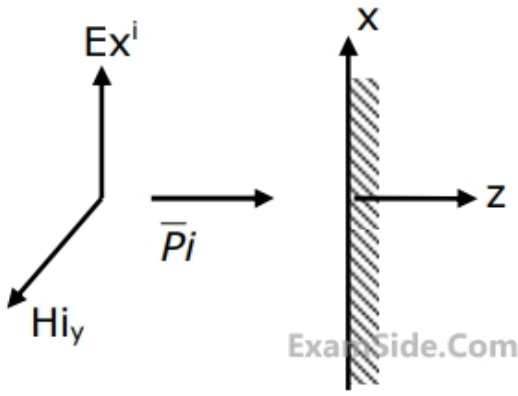
(C) neither a good conductor nor a good dielectric

(D) a good magnetic material

Q62. GATE ECE 1993

A plane wave is incident normally on a perfect conductor as shown in Fig. Here $E_x^i,\ H_y^i$ and $\overrightarrow{P^i}$ are electric field, magnetic field and Poynting vector respectively, for the incident wave. The reflected wave should have

Open solution



(A) $E_x^r = -E_x^i$ (B) $H_y^r = -H_y^i$

(C) $\overrightarrow{P^r} = -\overrightarrow{P^i}$ (D) $E_x^r = E_x^i$

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Q63. GATE ECE 1991

The electric field component of a uniform plane electromagnetic wave propagating in the Y-direction in a lossless medium will satisfy the equation

(A) $\frac{\partial^2 E_y}{\partial y^2} = \mu \in \frac{\partial^2 E_y}{\partial t^2}$

(B) $\frac{\partial^2 E_y}{\partial x^2} = \mu \in \frac{\partial^2 E_y}{\partial t^2}$

(C) $\frac{\partial^2 E_x}{\partial y^2} = \mu \in \frac{\partial^2 E_x}{\partial t^2}$

(D) $\frac{\sqrt{E_x^2+E_z^2}}{\sqrt{H_x^2+H_z^2}} = \sqrt{\mu / \in}$

Open solution

Q64. GATE ECE 1989

The skin - depth of copper at a frequency of $3GHz$ is 1 micron (10^{-6} metre). At $12GHz$, for a non - magnetic conductor whose conductivity is $1/9$ times that of copper, the skin $-$ depth would be

(A) $\sqrt{9 \times 4}$ microns

(B) $\sqrt{9/4}$ microns

(C) $\sqrt{4/9}$ microns

(D) $\frac{1}{\sqrt{9 \times 4}}$ microns

Open solution

Q65. GATE ECE 1988

In a good conductor the phase relation between the tangential components of electric field E_t and the magnetic field H_t is as follows

(A) E_t and H_t are in phase

(B) E_t and H_t are out of phase

(C) H_t leads E_t by 90^0

(D) E_t leads H_t by 45^0

Open solution

Q66. GATE ECE 1987

For an electromagnetic wave incident from one medium to a second medium, total reflection takes place when

(A) The angle of incidence is equal to the Brewster angle with E field perpendicular to the plane of incidence.

(B) The angle of incidence is equal to the Brewster angle with E field parallel to the plane of incidence.

(C) The angle of incidence is equal to the critical angle with the wave moving from the denser medium to a rarer medium.

(D) The angle of incidence is equal to the critical angle with the wave moving from a rarer medium to a denser medium.

Open solution

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Marks 5 8 questions

Q67. GATE ECE 2001

A medium has breakdown strength of 16 KV/m r.m.s. Its relative permeability is 1.0 and relative permittivity is 4.0. A plane electromagnetic wave is transmitted through the medium. Calculate the maximum possible power flow density and the associated magnetic filed.

Open solution

Q68. GATE ECE 1999

A plane wave in free space with

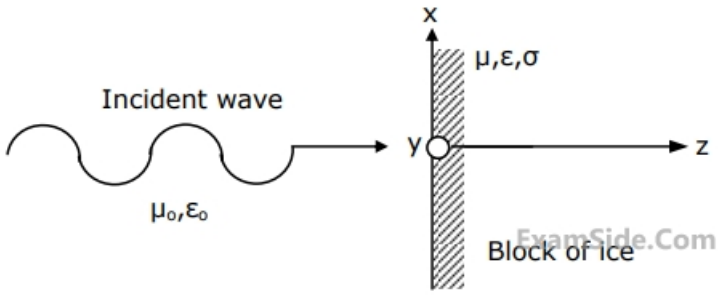
$\overrightarrow{E} = \left(\sqrt{\pi}\right)\left(10.0\,\hat{x} + 11.8\,\hat{y}\right)\exp\left[j\left(4\pi \times 10^8\,t - k\,z\right)\right]$

where $\hat{x}$ and $\hat{y}$ are unit vectors in the x- and y-directions respectively is incident normally on a semi-infinite block of ice as shown in Fig. For ice, $\mu = \mu_0,\ \ \sigma = 0$ and $\varepsilon = 9\varepsilon_0\,(1 - j0.001)$.

(a) Calculate the average power density associated with the incident wave.

(b) Calculate the skin depth in ice.

(c) Estimate the average power density at a distance of 5 times the skins depth in the ice block, measured from the interface.



Open solution

Q69. GATE ECE 1998

A plane wave with $\overrightarrow{E} = 10\,e^{j(\omega t - \beta z)}\ \overrightarrow{a}_y$ is incident normally on a thick plane conductor lying in the $X - Y$ plane. Its conductivity is $6 \times 10^6\ \ S/m$ and surface impedance is $5 \times 0^{-4} \angle 45^0 \Omega$. Determine the propagation constant and the skin depth in the conductor.

Open solution

Q70. GATE ECE 1998

The electric field vector of a wave is given as

$$\vec{E} = E_0 e^{j(\omega t + 3x - 4y)}\,\frac{8\vec{a}_x + 6\vec{a}_y + 5\vec{a}_z}{\sqrt{125}}\ V/m$$

Its frequency is 10 GHz.

(i) Investigate if this wave is a plane wave.
(ii) Determine its propagation constant, and
(iii) Calculate the phase velocity in y-direction.

Open solution

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Q71. GATE ECE 1997

A uniform plane wave is normally incident from air on an infinitely thick magnetic material with relative permeability 100 and relative permittivity 4 (sec in Fig.). The wave has an electric field of 1 V/meter (rms). Find the average Poynting vector inside the material.

Open solution

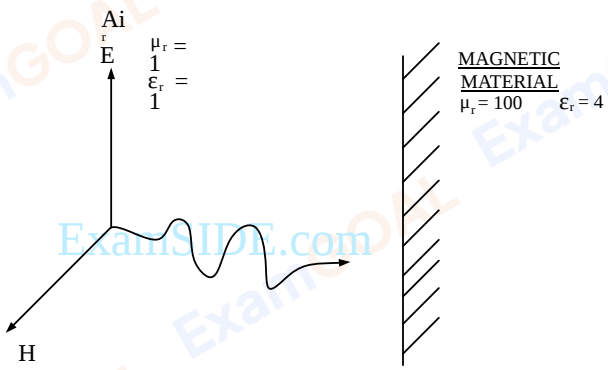


Fig.

Q72. GATE ECE 1996

A uniform plane wave having parallel polarization is obliquely incident on an air - dielectric interface as shown in Fig. If the wave has an electric field $E = 10\ V/m$, find
(i) The angle of incidence θ_i for which there is no reflection of the wave.
(ii) The surface charge density at the interface.

Open solution

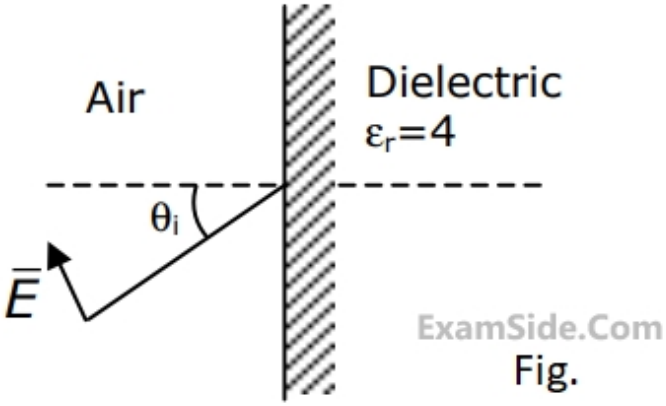


Fig.

Q73. GATE ECE 1994

A wave traveling in the $+Z$-direction, is the resultant of two linearly polarized components, $E_x = 3 \cos \omega t$ and $E_y = 2 \cos(\omega t + 45^\circ)$
Determine
(1) The axial ratio
(2) The angle between the major axis of the polarization ellipse and the $+x$-axis.

Open solution

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Q74. GATE ECE 1993

Open solution

Match the following descriptions with each of the diagrams given in Fig. Fields are near the interface, but on opposite sides of the boundary. Vectors are drawn to scale.

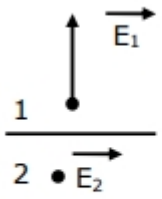
(a) Medium 1 and medium 2 are dielectrics with $\varepsilon_1 > \varepsilon_2$

(b) Medium 1 and medium 2 are dielectrics with $\varepsilon_1 < \varepsilon_2$

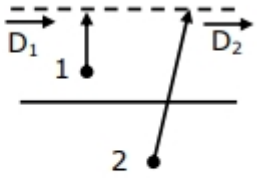
(c) Medium 2 is a perfect conductor

(d) Impossible

(e) Medium 1 is a perfect conduct

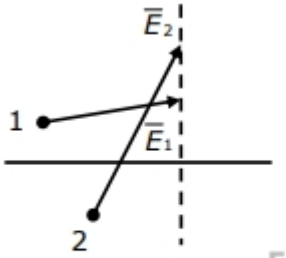


(i)

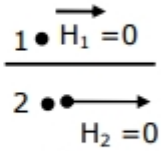


(ii)

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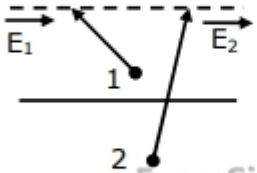


(iii)



(iv)

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(v)

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Marks 8 4 questions

Q75. GATE ECE 1992

A uniform plane wave traveling in free space along the +z direction and having its electric field along the$x$-direction, is normally incident on a thick brass sheet, infinite in extent (along the x-and y-directions). The electric field intensity of the wave is given by

$$E_x = 1225 \cos\left(5.89 \times 10^{10} t - \beta z\right) \quad V/m$$

.

Calculate the power per square metre that causes heating of the brass sheet, taking $\in_r = 1, \ \mu_r = 1$ and $\sigma = 1.649 \times 10^7$ mhos/metre for brass.

Open solution

Q76. GATE ECE 1991

A uniform plane electromagnetic wave traveling in free-space enters into a lossless medium at normal incidence. In the medium its velocity reduces by 50% and in free space sets up a standing wave having a reflection coefficient of - 0.125. Calculate the permeability and the permittivity of the medium.

Open solution

Q77. GATE ECE 1990

A 300 MHz plane E.M. wave is propagating in free space. The wave is incident normally on an infinite copper slab. For the transmitted wave in the copper slab, calculate the following
(a) Attenuation constant
(b) Phase constant
(c) Skin depth
(d) Phase velocity and
(e) Group velocity

for copper assume the following values
Conductivity $\sigma = 5.8 \times 10^7 \ mho/m$
Permeability $\mu = 4\pi \times 10^{-7} \ H/m$
Permeability $\varepsilon = \frac{1}{36\pi} \times 10^{-9} \ F/m$

Open solution

Q78. GATE ECE 1987

An electric field is given by $\overrightarrow{E} = \overrightarrow{a}_x \mathrm{A} \ \sin \beta y Sin \ \omega t$ is present in a charge free medium infinite in extent
(a) Determine the associated magnetic field.
(b) What type of wave does the above field represent?

Open solution

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Answer Key

Marks 1

1. **A**	2. **D**	3. **A, C**	4. **C**	5. **D**
6. **B**	7. **D**	8. **8.8**	9. **B**	10. **B**
11. **A**	12. **C**	13. **A**	14. **C**	15. **C**
16. **B**	17. **B**	18. **C**	19. **D**	20. **B**
21. **C**	22. **A**	23. **C**	24. **A**	25. **C**

Marks 2

26. **A**	27. **A, B, D**	28. **0.12 to 0.14**	29. **56.548**	30. **A**
31. **0**	32. **1**	33. **C**	34. **C**	35. **B**
36. **53.3**	37. **0.1**	38. **55.5**	39. **A**	40. **D**
41. **30**	42. **C**	43. **A**	44. **D**	45. **A**
46. **C**	47. **D**	48. **D**	49. **C**	50. **A**
51. **D**	52. **C**	53. **D**	54. **B**	55. **C**
56. **A**	57. **B**	58. **C**	59. **B**	60. **A**
61. **A**	62. **A, C**	63. **C, D**	64. **B**	65. **D**
66. **C**				

Marks 5

67. $1.36 \ \ MW/m^2$ **and** $85 \ \ A/m$	68. **(a)** $0.998 \ w/m^2$ **(b)** $159 \ m$ **(c)** $3 \times e^{-10} \ \ W/m^2$	69. $(21.2 \times 10^2) + j\,(21.2 \times 10^2)$ $0.47 \quad nm$	70. **(i) Plane wave** **(ii)** $j5.59$ **(iii)** 15.7×10^9 **m/sec**	71. $1.47\overrightarrow{a_r} \ \ mW/m^2$
72. **(i)** 63.4° **(ii)** $0.26 \ nc/m^2$	73. **(1)** 2.71 **(2)** 29.73°	74. **(A - iii), (B - ii), (C - i), (d - v) and (E - iv).**		

Marks 8

75. $0.25 \ W$	76. $\left(\frac{14}{9}\mu_0\right)H/m$ **and** $\frac{18}{7}\varepsilon_0 \quad F/m$	77. **(a)** $2.6 \times 10^5 \quad ncp/m$ **(b)** $2.6 \times 10^5 \quad r/m$ **(c)** $0.38 \times 10^{-2} \ mm$ **(d)** $7.2 \times 10^3 \ \ m/s$ **(e)** $1.25 \times 10^{13} \quad m/s$**.**	78. **(b) Standing wave**

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